



BITS Pilani presentation

BITS Pilani
Pilani Campus

Faculty Name
Faculty Department



EEE ZG512 / SS ZG512, EMBEDDED SYSTEM DESIGN

Lecture No. 05

Hardware / Software Design

Requirement Analysis

System Architecture – DFG & CFG

Component selection

Design process

Prototyping



Testing Stages

- Component Testing / Unit Testing (UT)
 - ✓ Electrical performance & Endurance
- Subsystem Level testing / Module Testing
 - ✓ Performance & Operation
- Integrated Testing
 - ✓ Functionality & Compliances

Interrupts

The interrupt mechanism allows devices to signal the CPU and to force execution of a particular piece of code (ISR – Interrupt Service Routine)

- The program counter's value is changed to point to an interrupt handler routine.

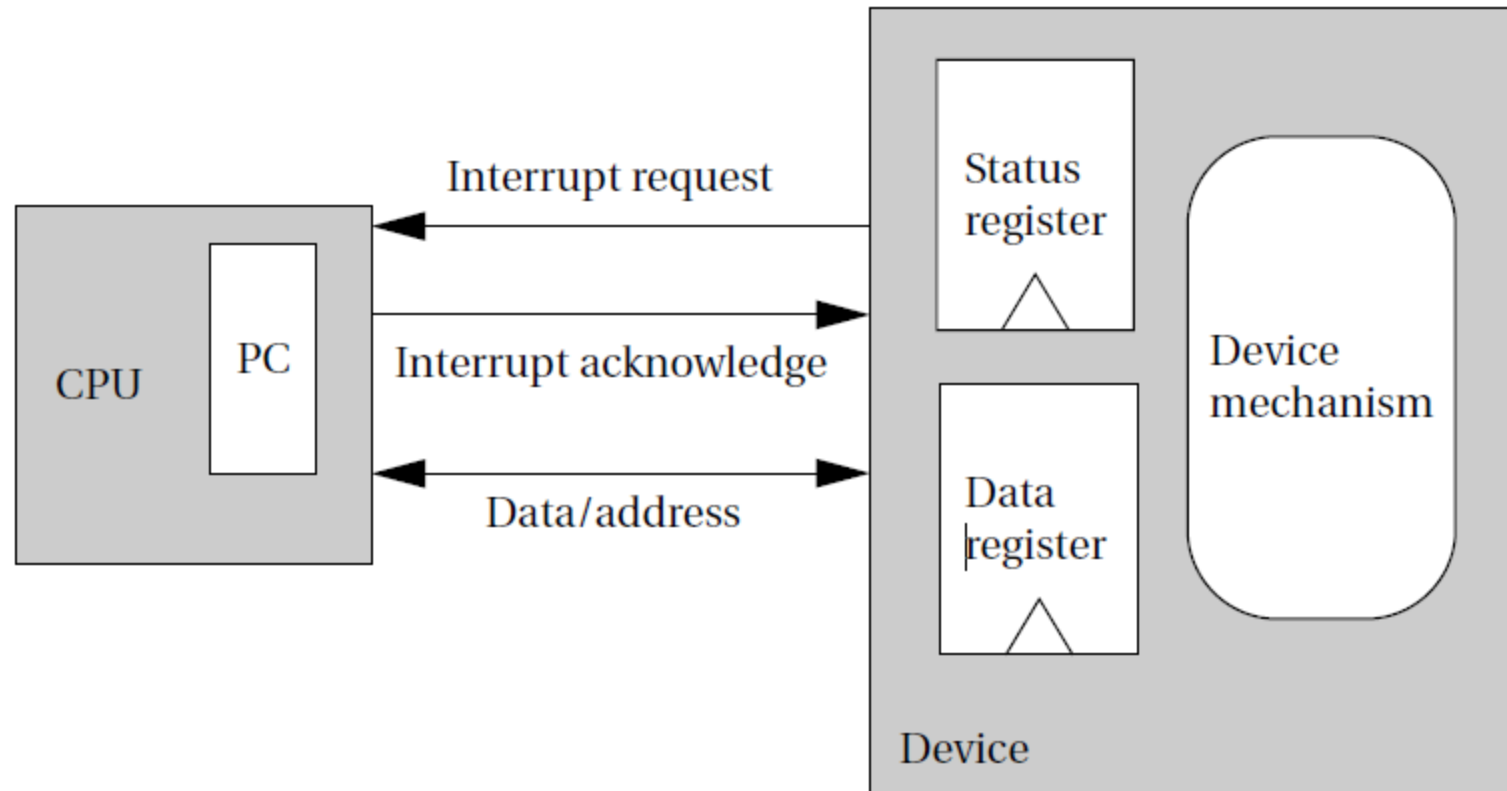


Interrupt Handling

- The I/O device asserts the interrupt request signal when it wants service from the CPU;
- The CPU asserts the interrupt acknowledge signal when it is ready to handle the I/O device's request.



Interrupt Mechanism



Interrupt Priorities & Vectors

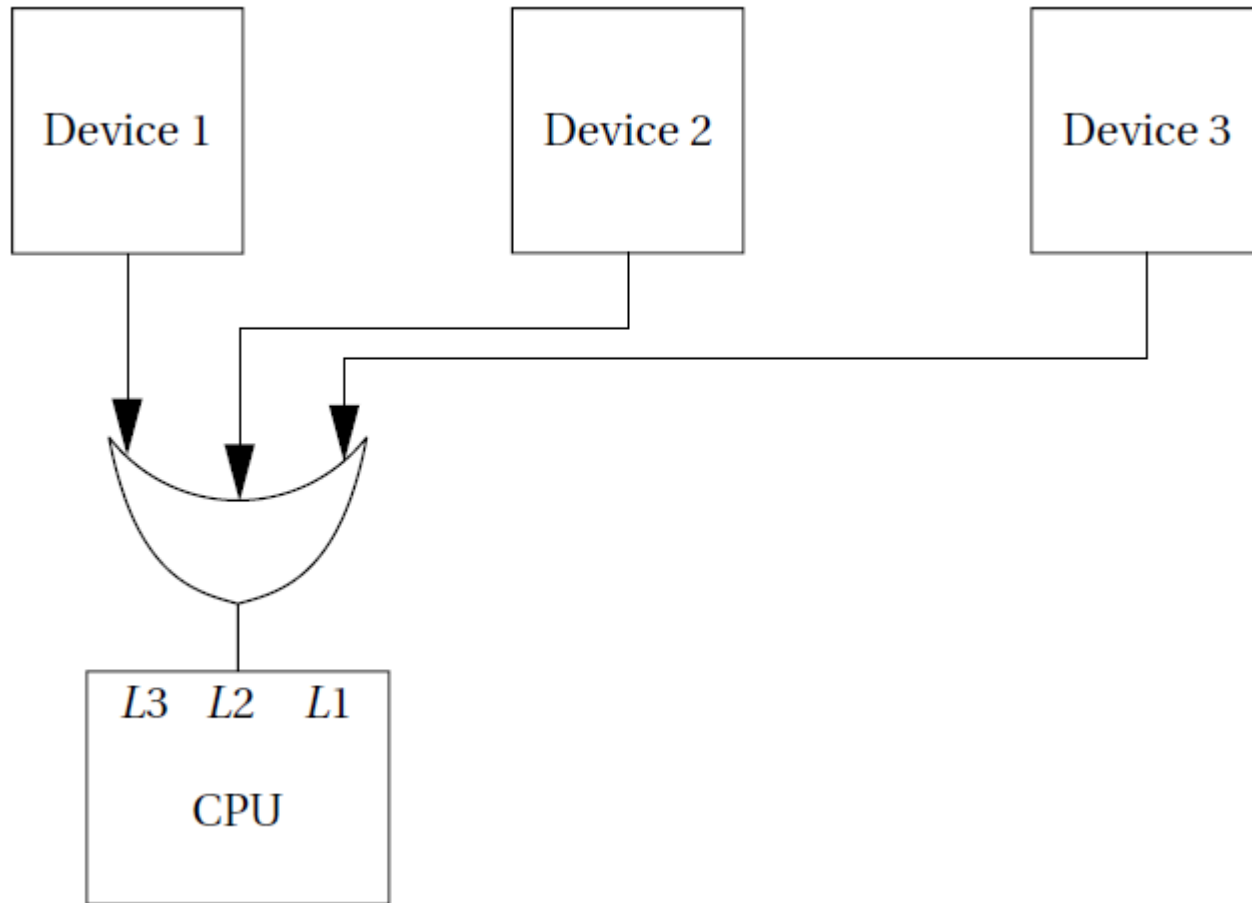
- Interrupt priorities allow the CPU to recognize some interrupts as more important than others.
- Interrupt vectors allow the interrupting device to specify its handler.
- The highest-priority interrupt is normally called the non-maskable interrupt (NMI).

Polling Mechanism

innovate

achieve

lead



Interrupts Overhead

- CPU: The CPU checks for pending interrupts at the beginning of an instruction.
- It answers the highest-priority interrupt, which has a higher priority than that given in the interrupt priority register.
- Device: The device receives the acknowledgment and sends the CPU its interrupt vector.
- CPU: Looks up the device handler address in the interrupt vector table (IVT).

Interrupts Overhead

- A subroutine-like mechanism is used to save the current value of the PC and possibly other internal CPU state, such as general-purpose registers.
- Software: It performs the required operations on the device.
- Then restores any saved state and executes the interrupt return instruction.
- CPU: The interrupt return instruction restores the PC and other automatically saved states.

Interrupts ARM7

Responding to an interrupt

- Saves the appropriate value of the PC to be used to return,
- Copies the CPSR into a saved program status register (SPSR),
- Forces bits in the CPSR to note the interrupt and forces the PC to the appropriate interrupt vector.

When leaving the interrupt handler, the handler should:

- Restore the proper PC value,
 - Restore the CPSR from the SPSR, and
 - Clear interrupt disable flags.
-

Worst-case Latency

Interrupt includes the following components:

- Two cycles to synchronize the external request,
- Up to 20 cycles to complete the current instruction,
- Three cycles for data abort and two cycles to enter the interrupt handling state.

This adds up to 27 clock cycles. The best-case latency is four clock cycles.

Timers



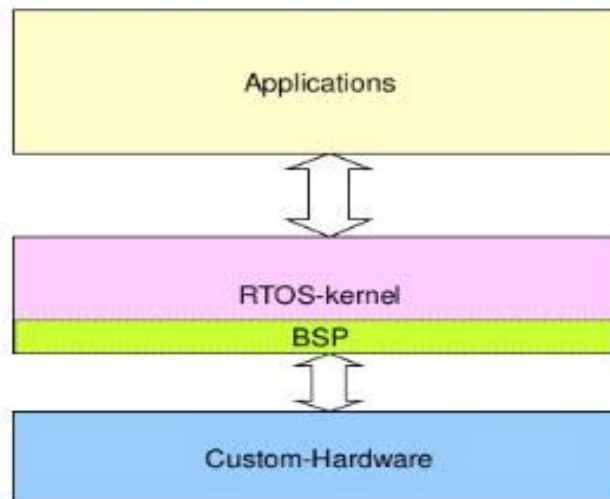
- Interval Timers: Timer has its count connected to a periodic clock signal to measure time intervals.
- Watchdog Timers: Resets the CPU if operation is not completed in stipulated time period.
- Brownout Timers /circuits: Restores the data in case of power failure to safeguard the system.

RTOS Structure



- RTOS is special type of OS designed to complete the tasks within time or deadline.
- Ex: FreeRTOS, μ COS etc...

Structure of RTOS



Tasks

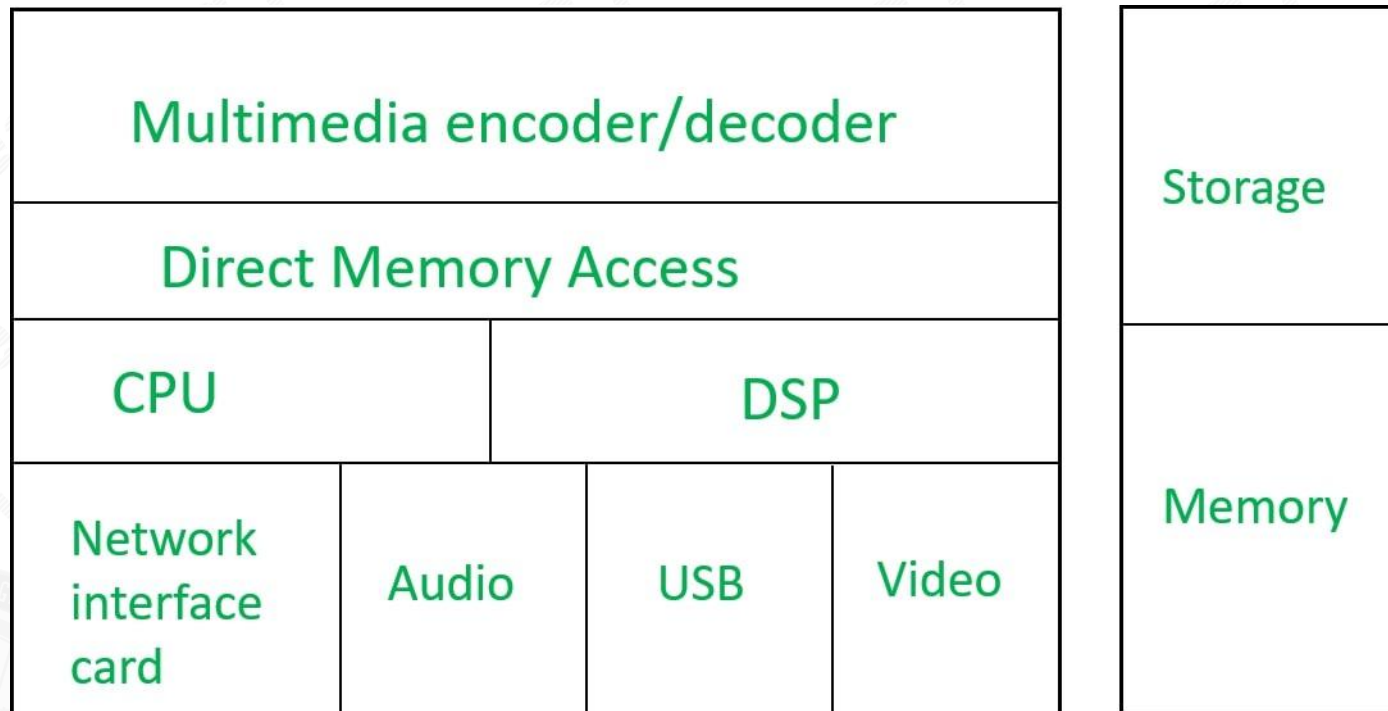
- Tasks: Task is a module with defined objective.
- Periodic: Triggered at regular time intervals.
- Dynamic: Aperiodic or sporadic tasks, usually event triggered.
- Critical: triggered on missed deadlines, hardware failures, resource allocation failed etc...
- Non-critical: Time management, information updating etc...



Introduction to SoC



Contains all the required components for the system operation.



END
of
Session – 5
Thank You

